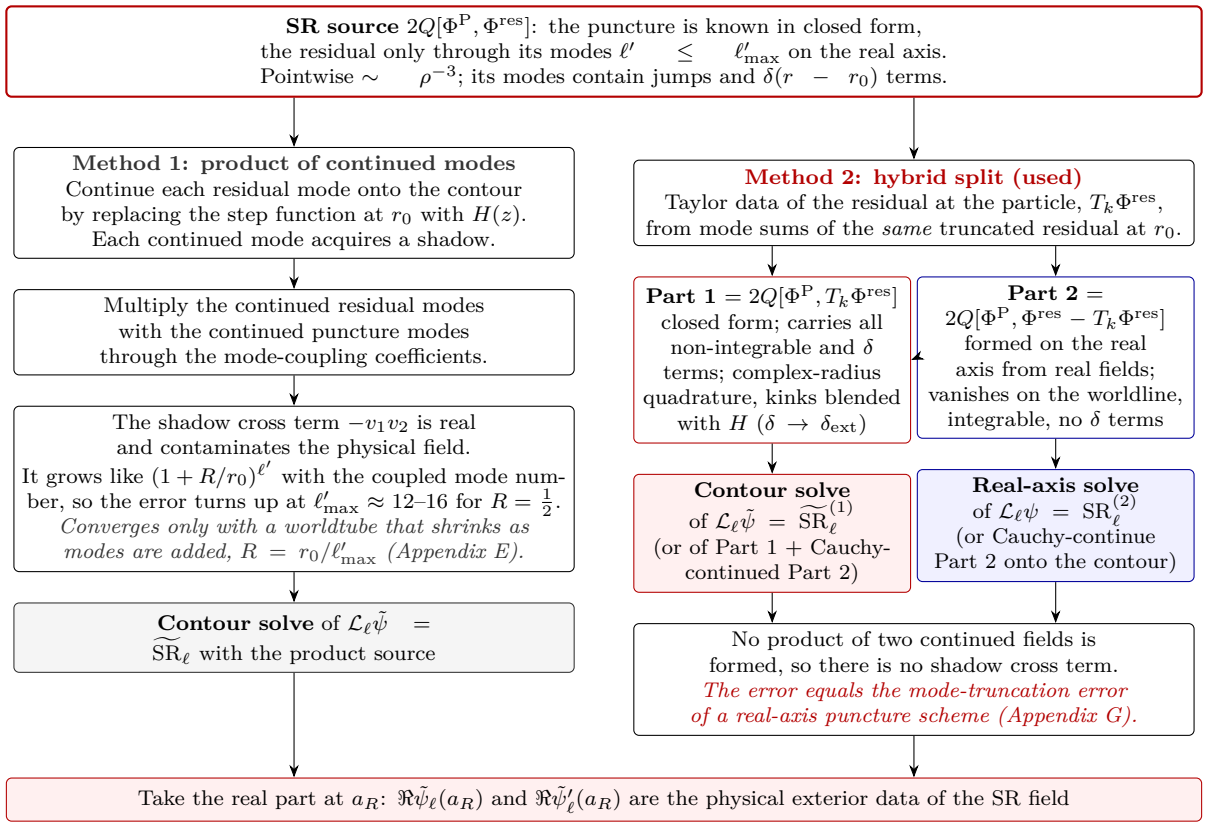

SR source $2Q[\Phi^{\rm P}, \Phi^{\rm res}]$: the puncture is known in closed form, the residual only through its modes $\ell' \leq \ell'_{\rm max}$ on the real axis. Pointwise $\sim \rho^{-3}$; its modes contain jumps and $\delta(r - r_0)$ terms.
Method 1: product of continued modes
Continue each residual mode onto the contour by replacing the step function at r_0 with $H(z)$. Each continued mode acquires a shadow.
Multiply the continued residual modes with the continued puncture modes through the mode-coupling coefficients.
The shadow cross term $-v_1 v_2$ is real and contaminates the physical field.
It grows like $(1 + R/r_0)^{\ell'}$ with the coupled mode number, so the error turns up at $\ell'_{\rm max} \approx 12$–$16$ for $R = \frac{1}{2}$.
Converges only with a worldtube that shrinks as modes are added, $R = r_0/\ell'_{\rm max}$ (Appendix E).
Contour solve of $\mathcal{L}_\ell \tilde{\psi} = \widetilde{\rm SR}_\ell$ with the product source
Method 2: hybrid split (used)
Taylor data of the residual at the particle, $T_k \Phi^{\rm res}$, from mode sums of the same truncated residual at r_0.
Part 1 $= 2Q[\Phi^{\rm P}, T_k \Phi^{\rm res}]$ closed form; carries all non-integrable and δ terms; complex-radius quadrature, kinks blended with H ($\delta \to \delta_{\rm ext}$)
Part 2 $= 2Q[\Phi^{\rm P}, \Phi^{\rm res} - T_k \Phi^{\rm res}]$ formed on the real axis from real fields; vanishes on the worldline, integrable, no δ terms
Contour solve of $\mathcal{L}_\ell \tilde{\psi} = \widetilde{\rm SR}^{(1)}_\ell$ (or of Part 1 + Cauchy-continued Part 2)
Real-axis solve of $\mathcal{L}_\ell \psi = {\rm SR}^{(2)}_\ell$ (or Cauchy-continue Part 2 onto the contour)
No product of two continued fields is formed, so there is no shadow cross term.
The error equals the mode-truncation error of a real-axis puncture scheme (Appendix G).
Take the real part at a_R: $\Re\tilde{\psi}_\ell(a_R)$ and $\Re\tilde{\psi}'_\ell(a_R)$ are the physical exterior data of the SR field